Class 11 Computer Science – Worksheet on Strings

Section A: Conceptual (2 × 2 = 4 Marks)

Q1. Strings are immutable in Python. Explain this property with an example.
Q2. Differentiate between the following with examples:

- isalpha() vs isalnum()
- find() vs rfind()

Section B: Output-Based Questions (5 × 2 = 10 Marks)

Q3. Predict the output:

```
s = "Python"
print(s[1:4])
print(s[-5:-2])
```

Q4.

```
s = "banana"
print(s.find("na"))
print(s.rfind("na"))
```

Q5.

```
msg = "HELLO world"
print(msg.lower())
print(msg.swapcase())
print(msg.title())
```

Q6.

```
s = "datascience"
```

```
print(s[::2])
print(s[::-1])
```

Q7.

```
text = "12345"
print(text.isdigit())
print(text.isalpha())
print(text.isalnum())
```

Section C: Error Correction (3 × 2 = 6 Marks)

Q8. Identify and correct errors:

```
s = "hello"
s[0] = "H"
print(s)
```

Q9. Debug the following:

```
word = "Python"
print(word.replace(P, p))
```

Q10. Correct and execute:

```
s = "apple"
print(s.append("s"))
```

Section D: Coding Questions (6 × 4 = 24 Marks)

Q11. Write a program to count the number of vowels, consonants, digits, and spaces in a string.

Q12. Write a program to input a string and check if it is a palindrome. Example: "madam" → Palindrome

Q13. Write a program to accept a string and display frequency of each character.

Q14. Write a program to input a string and print all words starting with a vowel.

Q15. Write a program to input a string and print:

- All uppercase letters
- All lowercase letters
- All digits

Q16. Write a program to input a sentence and display the longest word.

Section E: Application/Case-Based (2 × 6 = 12 Marks)

Q17. A password is valid if:

- At least 8 characters
- Contains at least one uppercase, one lowercase, one digit, and one special character
  Write a program to check whether a password entered is valid.

Q18. A paragraph is entered as input (multi-line string). Write a program to display:

- Number of lines
- Total number of words
- Frequency of each word